One-Sample t-Test

2026-04-17

Introduction

The one-sample t-test compares the mean of a single continuous variable to a pre-specified reference value. It answers questions of the form "does the average differ from some benchmark?" – average IQ differs from 100, mean biomarker level differs from a reference, average weight change differs from zero in a pre-post design (via paired differences).

Prerequisites

t-distribution, the concept of a sampling distribution.

Theory

Given iid observations $X_1, \ldots, X_n$ from $\mathcal{N}(\mu, \sigma^2)$, the test of

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0$$

uses the statistic

$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad \text{under } H_0.$$

The sample SD s replaces the population σ; the distribution adjusts from normal to t because this replacement adds variability. The degrees of freedom are $n-1$.

Confidence interval for μ: $\bar{X} \pm t_{1-\alpha/2,n-1} \cdot s/\sqrt{n}$.

Effect size: Cohen's $d = (\bar{X} - \mu_0)/s$.

Assumptions

- Independent observations.
- Normal distribution (or n large enough for CLT to kick in).
- No extreme outliers affecting the mean.

R Implementation

```r
library(effectsize); library(broom)
set.seed(2026)

# Simulated IQ-test scores; is the mean different from 100?
iq <- rnorm(45, mean = 103.5, sd = 14)

t.test(iq, mu = 100) |> tidy()
```

```r
# Effect size
cohens_d(iq, mu = 100)

# Check assumptions
shapiro.test(iq)
boxplot(iq, main = "Sample distribution", col = "#2A9D8F")
```

Output & Results

```
# A tibble: 1 x 8
  estimate statistic p.value parameter conf.low conf.high method  alternative
     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>   <chr>
1     104.      1.82   0.076        44     98.9      109.     One ... two.sided

Cohen's d |         95% CI
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0.27      | [-0.03, 0.57]
```

The sample mean is 104, not significantly different from 100 at $\alpha = 0.05$ ($t(44) = 1.82$, $p = 0.076$), with a small-to-medium effect ($d = 0.27$).

Interpretation

In a manuscript: "The mean IQ was 104.0 (SD 14.0). The 95 % CI for the mean (98.9 to 109.3) includes the reference value of 100; a one-sample t-test did not reject the null of equal mean (t(44) = 1.82, p = 0.076, Cohen's d = 0.27)."

Practical Tips

- For $n \geq 30$, the CLT makes the t-test fairly robust to non-normality.
- For small n with clear non-normality, use the Wilcoxon signed-rank test (one-sample version).
- Power for the one-sample t-test depends on effect size and n; use `power.t.test(type = "one.sample")`.
- The reference value μ_0 must be pre-specified; post-hoc selection invalidates the p-value.
- Pair with a 95 % CI for the mean – it is more informative than the p-value alone.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds

- Pearson and Spearman correlation
- Non-parametric tests